Model-Guided Design of Porous Polyimide/Boron Nitride Films for Low-Permittivity, Thermally Conductive Packaging

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Abstract

Packaging films for high-frequency and power electronics must combine a low dielectric constant with useful thermal conductivity and dimensional stability, but the strategies that improve one property usually degrade another. Porous polymers reach the lowest permittivities at the expense of heat transport and strength, and thermally conductive fillers raise permittivity. Here we report porous polyimide (PI)/hexagonal boron nitride (h-BN) films, made by non-solvent induced phase separation (NIPS) over 0–70 wt% h-BN and restructured by hot-pressing, in which porosity and filler loading are treated as two separate design variables. We measure the dielectric, thermal, thermomechanical, and mechanical properties of all ten films and describe them with a single three-phase Lewis–Nielsen model in which each film is a PI, h-BN, and air system. The fitted pore shape factor ($A_{\mathrm{pore}} = 0.08$) fingerprints the bicontinuous NIPS sponge, and a single-parameter thermal calibration reproduces the conductivity data to within $0.03\,\mathrm{W\,m^{-1}\,K^{-1}}$. The model identifies 70 wt% h-BN at 50–60% porosity as the most favorable design region: the non-pressed film reaches $\varepsilon_r = 1.28$ and $\lambda = 0.31\,\mathrm{W\,m^{-1}\,K^{-1}}$, and the hot-pressed film combines $\varepsilon_r = 1.62$ and $\lambda = 0.28\,\mathrm{W\,m^{-1}\,K^{-1}}$ with a CTE of $10.1\,\mathrm{ppm\,K^{-1}}$. These results establish the pore shape factor as a microstructural design parameter for porous ceramic-filled polymer films and provide a calibrated basis for selecting composition and porosity in NIPS-derived packaging films.

Keywords: Polymer-matrix composites, Hexagonal boron nitride, Porous film, Lewis–Nielsen model, Dielectric properties, Thermal conductivity

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1. Introduction

High-density microelectronics, millimeter-wave fifth-generation (5G) communication, and electric-vehicle power electronics place three demands on a polymer packaging film at once: a low dielectric constant ($\varepsilon_r < 2.5$) to limit signal delay and cross-talk at GHz–THz frequencies; a thermal conductivity above $\sim 0.2\,\mathrm{W\,m^{-1}\,K^{-1}}$ to dissipate heat fluxes that now exceed $100\,\mathrm{W\,cm^{-2}}$ in advanced packages; and a coefficient of thermal expansion (CTE) below $\sim 20\,\mathrm{ppm\,K^{-1}}$, matched to copper ($\sim 17\,\mathrm{ppm\,K^{-1}}$) and silicon ($\sim 3\,\mathrm{ppm\,K^{-1}}$) so that thermal cycling does not delaminate the stack [1, 2, 3]. No existing polymer meets all three without modification, and the strategies that improve one property typically degrade another. This is the central difficulty of multifunctional packaging design.

Aromatic polyimide (PI) has been the dielectric substrate of choice for decades because of its thermal stability (decomposition above 500 °C), strength, chemical resistance, and electrical insulation [4]. The canonical PMDA–ODA system used here has a dielectric constant of ∼3.4–3.8, a thermal conductivity of $\sim 0.12\,\mathrm{W\,m^{-1}\,K^{-1}}$, and a CTE of ~ 40–$60\,\mathrm{ppm\,K^{-1}}$ in dense form, none of which meets the requirements above [5]. Molecular strategies that lower the dielectric constant, such as fluorinated monomers, bulky pendant groups, and low-polarizability diamines, reach $\varepsilon_r < 3.0$ but need specialized monomers and do nothing for thermal conductivity [6, 7].

Porosity offers a complementary route. In non-solvent induced phase separation (NIPS), a poly(amic acid) solution in NMP is cast and immersed in a non-solvent such as acetone, water, or an alcohol; rapid solvent exchange drives spinodal decomposition or nucleation and growth, and the resulting bicontinuous sponge survives thermal imidization [8, 9, 10]. Because air has the lowest dielectric constant of any material, large air fractions lower the permittivity well below what molecular design alone achieves: NIPS-derived PI and related engineering polymers reach $\varepsilon_r = 1.51$–2.48 at porosities up to 75%, tunable through polymer concentration, non-solvent, and immersion conditions [7, 11, 12]. Unlike aerogel and freeze-drying routes, NIPS needs neither supercritical drying nor cryogenic equipment and is compatible with roll-to-roll processing [13].

Hexagonal boron nitride (h-BN) addresses the thermal and dimensional shortfalls of porous PI. It is a wide-bandgap ceramic ($E_g \approx 6\,\mathrm{eV}$) whose layered structure gives a strongly anisotropic thermal conductivity, up to $400\,\mathrm{W\,m^{-1}\,K^{-1}}$ in-plane and ∼**[30]** $\mathrm{W\,m^{-1}\,K^{-1}}$ through-plane, among the highest of any electrical insulator [14]. Its dielectric constant (∼6.0; $\varepsilon_\perp = 7.04$, $\varepsilon_\parallel = 4.95$) is well below that of aluminum nitride ($\varepsilon_r \approx 9$) or silicon carbide ($\varepsilon_r \approx 10$), so it is the filler of choice when dielectric performance must be preserved [15]. Its in-plane CTE ($\sim 1.5\,\mathrm{ppm\,K^{-1}}$) and modulus (∼200–700 GPa) also make it an effective dimensional stabilizer [16, 17].

Dense PI/h-BN films reach 0.5–$6.2\,\mathrm{W\,m^{-1}\,K^{-1}}$ at 10–60 wt% h-BN, but at $\varepsilon_r = 3.5$–5.0 [18, 19, 20]. Yang et al. [21] combined a high-internal-phase Pickering emulsion (HIPPE) template with hot-pressing to obtain porous h-BN/PI films with low ε_r and high thermal diffusivity. The spherical, emulsion-templated pores of HIPPE films are, however, geometrically different from the bicontinuous sponge of NIPS, and the two architectures should contribute differently to composite properties; this difference has not been quantified in a single model. Porous PI films also retain their dielectric properties at elevated temperature [22], and electrospinning–hot-press routes give dense h-BN/PI composites of high thermal stability [15]. What is missing is a quantitative model that relates dielectric constant and thermal conductivity to both h-BN loading and porosity, returns physically interpretable microstructure parameters, and yields design guidance for NIPS-derived porous PI/h-BN films.

The Lewis–Nielsen (LN) model [23, 24], a modification of the Halpin–Tsai equations, provides this capability. It predicts the effective scalar transport property of a particulate composite from the filler volume fraction, the filler–matrix property contrast, a filler shape factor (the Einstein coefficient A), and a maximum packing fraction $\phi_{\max}$. Applied sequentially, first mixing PI with h-BN and then introducing air pores, it becomes a three-phase model that takes measured porosities and loadings as inputs and returns parameters that characterize the h-BN platelet geometry and the NIPS pore architecture [25]. The Turner model [26] plays the corresponding role for CTE, and the Gibson–Ashby scaling [27] for the strength of open-cell porous solids.

Here we report a combined experimental and modeling study of NIPS-derived porous PI/h-BN films across 0–70 wt% h-BN in both non-pressed (NP) and hot-pressed (HP) states. The central contribution is a calibrated three-phase LN framework that describes the measured dielectric constant and thermal conductivity with one shared microstructural parameterization, extracts interpretable composite parameters, and generates design maps for simultaneous property optimization. The fitted pore shape factor $A_{\text{pore}} = 0.08$ is a microstructure fingerprint that distinguishes NIPS sponge pores from the spherical pores of HIPPE composites; a deliberately parsimonious single-parameter thermal calibration avoids the over-fitting common to multi-parameter composite models; and the calibrated model prescribes the porosity needed to meet dual-property specifications. We evaluate the framework against ten films to show that porosity and pore geometry, not filler geometry, govern the dielectric response, whereas thermal conductivity is gated by filler loading through the approach to maximum packing.

2. Materials and methods

2.1. Materials

Pyromellitic dianhydride (PMDA, >98%, Tokyo Chemical Industry, Japan), 4,4′-oxydianiline (ODA, >98%, TCI), and hexagonal boron nitride (h-BN, 99.5%, mean particle diameter ∼70 nm, MK Impex Corp., Canada) were used as received. N-Methyl-2-pyrrolidone (NMP, 99.5%) and acetone (99.5%) were purchased from Duksan Pure Chemicals, Korea.

2.2. PAA/h-BN dispersion synthesis

ODA (10 mmol, 2.0024 g) was dissolved in NMP (23.707 g) at room temperature, and h-BN was dispersed in the solution by probe sonication (30 min). PMDA (10 mmol, 2.1812 g) was added portionwise with stirring for 24 h at room temperature to give a viscous poly(amic acid) (PAA)/h-BN solution at equimolar

PMDA:ODA. h-BN was added at 0, 10, 30, 50, or 70 wt% of the total PAA + h-BN solids while the PAA concentration in NMP was held at 15 wt% for all compositions, so that the phase-separation thermodynamics of the polymer solution were the same across the series and h-BN loading was the only compositional variable.

2.3. NIPS film fabrication and thermal imidization

The PAA/h-BN solutions were spin-coated onto soda-lime glass (1000 rpm, 10 s); the high viscosity of the 15 wt% PAA solution and the short spin time leave a wet film of several hundred micrometres, which after NIPS and imidization gives free-standing porous films of 52–172 µm (Figure 1). The coated substrates were immersed immediately in 20 mL of acetone at room temperature for 2 h. The solidified films were imidized at 200 °C for 2 h and 350 °C for 1 h in air and peeled from the glass to give the non-pressed (NP) films. FT-IR (Figure S8) confirmed complete imidization: imide C=O stretching at 1780 and 1720 cm^{-1}, C−N stretching at 1380 cm^{-1}, and no residual amide carbonyl band at ∼1660 cm^{-1}.

2.4. Hot-press densification

A second set of NP films was densified between polished steel plates in a hydraulic press (250 °C, 400 MPa, 30 min), following the protocol of our earlier work on NIPS-derived porous PI composites [28], to give the hot-pressed (HP) films. Because 250 °C lies well below the glass transition of PMDA–ODA PI (> 400 °C [28]), densification proceeds by thermally assisted plastic collapse of the glassy pore walls rather than by viscous flow. This is consistent with the partial, rather than complete, loss of porosity and with the resistance of the rigid h-BN network to compaction (Section 4.1). NP and HP films of the same composition were characterized side by side.

2.5. Characterization

Tensile strength was measured on a universal testing machine (5 mm min^{-1}; 10 mm wide, 50 mm gauge length, thickness by micrometer; ASTM D882). Dielectric constant was measured at 100 kHz with an LCR meter (**[E4980A, Keysight]**; parallel-plate contact electrodes, **[20]** mm diameter) on films **[without]** sputtered electrodes. The HP 30 wt% film was also measured independently at KOPTRI (Korea Polymer Testing & Research Institute; ASTM D150, 100 kHz to 13 MHz, Agilent LCR, G10 electrode, 23 ± 2 °C, 45 ± 5% RH); the two laboratories are compared in Section 4.2. Thermal diffusivity (α, laser flash) and specific heat (C_p, DSC, ASTM E1269, 5 °C min^{-1}, 0–300 °C) were measured at Korea University of Technology and Education (KOREATECH); bulk density (ρ) was obtained from film mass and optically measured dimensions, and $\lambda = \alpha\rho C_p$. CTE was measured by TMA (Q400 EM, TA Instruments; tension mode, 10 °C min^{-1}, N_2 100 mL min^{-1}, preload 0.05 N, 100–200 °C; Kyung Hee University). Open porosity was measured by mercury intrusion (PM33GT, Quantachrome; Cooperative Equipment Center, **[Yonsei University]**), and total porosity was calculated as $\phi_{total} = 1 - \rho/\rho_{solid}$ with ρ_{solid} from the PI and h-BN densities and the solid-phase composition (Section 3.1); the two are compared in Section 4.1. Microstructure and composition were examined by FE-SEM/EDS (JEOL-7800F and JEOL-7001F), and FT-IR spectra were recorded on a Jasco FT/IR-460 Plus (600–4000 $\mathrm{cm^{-1}}$).

Unless stated otherwise, each property was measured on one specimen per composition and processing state ($n = 1$), so the reported values carry the measurement uncertainty only. Representative uncertainties are ±5% in α, ±3% in C_p, and ±8% in ρ (limited by thickness), which propagate to about ±10% in λ; ±[0.05] in ε_r (about ±3%); and ±[1] ppm $\mathrm{K^{-1}}$ in CTE. These bounds set how far individual residuals can be interpreted in Sections 4.2 and 4.3.

3. Modeling framework

3.1. Sequential three-phase Lewis–Nielsen model

We treat each film as a three-component system of PI matrix (property k_m), h-BN filler (k_f), and air pores ($k_{\text{pore}} \approx k_{\text{air}}$). The term "three-phase Lewis–Nielsen model" has been used before for a matrix–filler–interphase construction in which the third phase is the interfacial polymer layer around the nanofiller [29]; here the third phase is the NIPS pore network. Following the sequential mixing approach [25], step 1 combines PI and h-BN into a dense composite,

$$\frac{k_{\text{dense}}}{k_m} = \frac{1 + A_{\text{BN}}\, B\, \phi_{\text{BN}}}{1 - B\, \psi\, \phi_{\text{BN}}}, \tag{1}$$

where

$$B = \frac{k_f/k_m - 1}{k_f/k_m + A_{\text{BN}}}, \qquad \psi = 1 + \frac{1 - \phi_{\max}}{\phi_{\max}^2}\, \phi_{\text{BN}}, \tag{2}$$

and step 2 introduces the pores into that composite:

$$\frac{k_{\text{final}}}{k_{\text{dense}}} = \frac{1 + A_{\text{pore}}\, B_p\, \phi_p}{1 - B_p\, \psi_p\, \phi_p}. \tag{3}$$

The LN model was formulated for elastic moduli and thermal conductivity [23, 24], but it applies without modification to the dielectric constant: permittivity and thermal conductivity are scalar transport properties governed by the same Laplace-type equation, $\nabla\cdot(k\nabla u) = 0$, so every effective-medium result transfers under $k \leftrightarrow \varepsilon$. In the dilute limit for spheres the LN expression reduces to Maxwell–Garnett, of which it is a generalization through A and ψ. Its advantage over the standard dielectric mixing rules (Lichtenecker, Maxwell–Garnett, Bruggeman) for the present system is twofold: it describes ε_r and λ with one shared microstructural parameterization, and it is the only one of these formalisms with an explicit pore-geometry parameter (A_{pore}) and a packing-limit divergence, the two features on which our conclusions rest. A benchmark against the zero-parameter mixing rules (Supplementary Material, Section S3.5) confirms that the calibrated LN model outperforms all of them, by the largest margin (RMSE 0.26 vs. 0.55–0.90 over the 30–70 wt% films) at moderate porosity.

The same equations are used for ε_r and λ. The h-BN weight fraction was converted to a volume fraction in the solid (PI + h-BN) phase with $\rho_{\mathrm{PI}} = 1.42\,\mathrm{g\,cm^{-3}}$ and $\rho_{\mathrm{BN}} = 2.29\,\mathrm{g\,cm^{-3}}$, and the measured open porosity was used as ϕ_p in Eq. 3; the relation between open and total porosity is examined in Sections 4.1 and 4.2. The full derivation, parameter tables, and Python implementation are given in Sections S2–S5.

3.2. Physical basis of the Einstein coefficient A

The Einstein coefficient A is the central microstructure parameter of the model. It originates in the classical treatment of field or flux distortion around a single inclusion in an infinite matrix [23, 24], and the three-phase system needs two of them: A_{BN} for the h-BN filler and A_{pore} for the pore network.

For platelets, A depends on the aspect ratio $\alpha = D/t$ and on orientation relative to the field. The nominal aspect ratio of the 70 nm h-BN would give $A_{\mathrm{rand}} \sim 4$–14, but three factors reduce the effective value: (i) probe sonication fragments edges and partially delaminates the platelets, lowering α toward 5–7; (ii) 70 nm particles readily stack face to face into more equiaxed inclusions; and (iii) the rapid NIPS exchange freezes the particles in random orientations. For randomly oriented platelets of $\alpha \approx 5$–7, $A_{\mathrm{rand}} \approx 4$–6 and $\phi_{\max} \approx 0.55$–0.65 [23, 24, 25]; we adopt these values for the thermal model ($A = 5.5$, $\phi_{\max} = 0.60$; Section 3.4). The dielectric fit returns $A_{\mathrm{BN}} = 0.68$, below the isolated-sphere value of 1.5, but the sensitivity analysis (Section 4.6, Figure S1a) shows that ε_r is nearly insensitive to A_{BN} over 0.3–6.0: the pore term (step 2, Eq. 3) dominates the dielectric response, so the dielectric A_{BN} is weakly determined and its precise value carries little physical weight.

For the pores, A measures how effectively the pore phase shields the matrix from the field. Isolated spherical pores give $A = 1.5$. In the bicontinuous NIPS sponge, the PI skeleton provides continuous conduction paths through the pore space, so the field disruption per unit porosity is much smaller. The fitted $A_{\mathrm{pore}} = 0.08$ quantifies this: at $\phi_p = 0.49$ (NP 30 wt%), spherical pores would give $\varepsilon_r \approx 2.22$ whereas $A_{\mathrm{pore}} = 0.08$ gives 1.71, and the difference of 0.51 is the signature of the sponge architecture. A_{pore} therefore serves as a microstructure fingerprint that separates NIPS sponge pores from the spherical Pickering-emulsion (HIPPE) pores of Yang et al. [21], for which $A \sim 1.2$–1.5 is expected.

3.3. CTE and mechanical models

CTE was predicted with the Turner model [26], $\mathrm{CTE}_c = (\sum_i \phi_i E_i \alpha_i)/(\sum_i \phi_i E_i)$, using $E_{\mathrm{BN}} = 200\,\mathrm{GPa}$ and $\alpha_{\mathrm{BN}} = 1.5\,\mathrm{ppm\,K^{-1}}$ [16], $E_{\mathrm{PI}} = 3.5\,\mathrm{GPa}$ [5], and for α_{PI} the value measured on the neat porous PI HP film ($41.92\,\mathrm{ppm\,K^{-1}}$, TMA) rather than the dense-PI literature value of $\sim 60\,\mathrm{ppm\,K^{-1}}$, because the sponge architecture changes the effective expansion of the PI phase. The linear rule of mixtures (ROM) is shown for comparison. As discussed in Section 4.4, the Turner model underpredicts and the ROM overpredicts the measured CTE at 10–70 wt% h-BN because the porous skeleton violates the continuous load-path assumption of the Turner model: platelets embedded in discontinuous thin PI ligaments cannot constrain the matrix as they would in a dense co-continuous composite, so the measured values fall between the two bounds. Tensile strength was modeled with the LN composite modulus and the Gibson–Ashby open-cell scaling, $\sigma \propto E_{\mathrm{dense}}(1-\phi_p)^2$ [27].

3.4. Parameter estimation and model validation

Dielectric model. We fitted four parameters (A_{BN}, $\phi_{\mathrm{max,BN}}$, A_{pore}, $\phi_{\mathrm{max,pore}}$) to the ten ε_r measurements (five compositions × two states) by multi-start Nelder–Mead minimization of the squared residuals (16 starts; Eq. S8). Both packing parameters converge to the upper bound (0.99) and are effectively inactive: fixing $\phi_{\mathrm{max,BN}}$ anywhere in 0.50–0.99 changes the RMSE by only 0.03 (Table S11). The dielectric model is therefore governed by two to three active parameters fitted to ten points.

Thermal model. When all three thermal parameters ($\lambda_{\mathrm{BN,eff}}$, $A_{\mathrm{BN},\lambda}$, $\phi_{\mathrm{max,BN},\lambda}$) are fitted freely, the optimizer converges to a degenerate solution ($A \to 700$, $\phi_{\mathrm{max}} \to 0.10$): A and ϕ_{max} are strongly correlated in Eq. 1, the data cannot separate them, and the fit is good but meaningless (Section S4.2). We therefore fixed $A_{\mathrm{BN},\lambda} = 5.5$ and $\phi_{\mathrm{max,BN},\lambda} = 0.60$, the literature values for randomly oriented platelets of $\alpha \approx 5$–7 [23, 24, 25] and consistent with the sonication-modified particle geometry of Section 3.2, and fitted only $\lambda_{\mathrm{BN,eff}}$. The result, $\lambda_{\mathrm{BN,eff}} = 6.81\,\mathrm{W\,m^{-1}\,K^{-1}}$ with RMSE 0.027 (NP) and $0.030\,\mathrm{W\,m^{-1}\,K^{-1}}$ (HP), is statistically indistinguishable from the unconstrained fit (0.027, 0.023), and every reported parameter is physically interpretable at a 7:1 data-to-parameter ratio. With $\phi_{\mathrm{max}} = 0.60$, the 70 wt% composition ($\phi_{\mathrm{BN}} = 0.591$) sits at 99% of maximum packing, so the packing term ψ in Eq. 1 produces the sharp conductivity rise seen at this loading; this is the LN representation of the approach to particle contact.

Validation. The calibrated dielectric model is compared with an independent inter-laboratory measurement in Section 4.2. Beyond that comparison, its credibility rests on three points: (i) the parsimony of the calibration (two to three active dielectric parameters, one thermal parameter); (ii) the shape-factor sensitivity analysis (Section 4.6), which shows that the conclusions do not depend on the weakly determined parameters; and (iii) the agreement of the model λ envelope with the measured 70 wt% films (Section 4.8, Figure 4d).

Table 1: Three-phase Lewis–Nielsen parameters: estimation protocol and physical interpretation.

Parameter	Symbol	Value	Status	Physical interpretation
h-BN shape factor (ε_r)	A_{BN}	0.68	Fitted (weakly det.)	ε_r is pore-dominated; A_{BN} carries little weight (Figure S1a)
h-BN max. packing (ε_r)	$\phi_{\mathrm{max,BN}}$	0.99	Boundary (inactive)	RMSE changes $\leq$0.03 over 0.50–0.99 (Table S11)
Pore shape factor	A_{pore}	0.08	Fitted	Bicontinuous NIPS sponge; cf. $A = 1.5$ for isolated spheres
Pore max. packing	$\phi_{\mathrm{max,pore}}$	0.99	Boundary	NIPS sponge supports up to 91% open porosity
h-BN shape factor (λ)	$A_{\mathrm{BN},\lambda}$	5.5	Fixed (literature)	Randomly oriented platelet, effective $\alpha \approx 5$–7
h-BN max. packing (λ)	$\phi_{\mathrm{max,BN},\lambda}$	0.60	Fixed (literature)	Random platelet packing; $\phi_{\mathrm{BN}}(70\,\mathrm{wt\%}) = 0.59 \to 99\%$ of ϕ_{max}
Eff. h-BN conductivity	$\lambda_{\mathrm{BN,eff}}$	$6.81\,\mathrm{W\,m^{-1}\,K^{-1}}$	Fitted (single)	Well below the bulk through-plane value; Kapitza resistance at 70 nm

4. Results and discussion

4.1. Film morphology in the NP and HP states

NIPS with acetone gives a bicontinuous sponge with 1–10 µm pores in every NP film, with the h-BN platelets embedded in the PI cell walls (Figure 1; photographs and top-view micrographs in Figures S6 and S7). Open porosities range from 48.8% (30 wt%) to 91.4% (10 wt%). The high value at 10 wt% may reflect a nucleation effect, with the h-BN particles acting as additional phase-separation nuclei that give finer, more uniformly distributed pores.

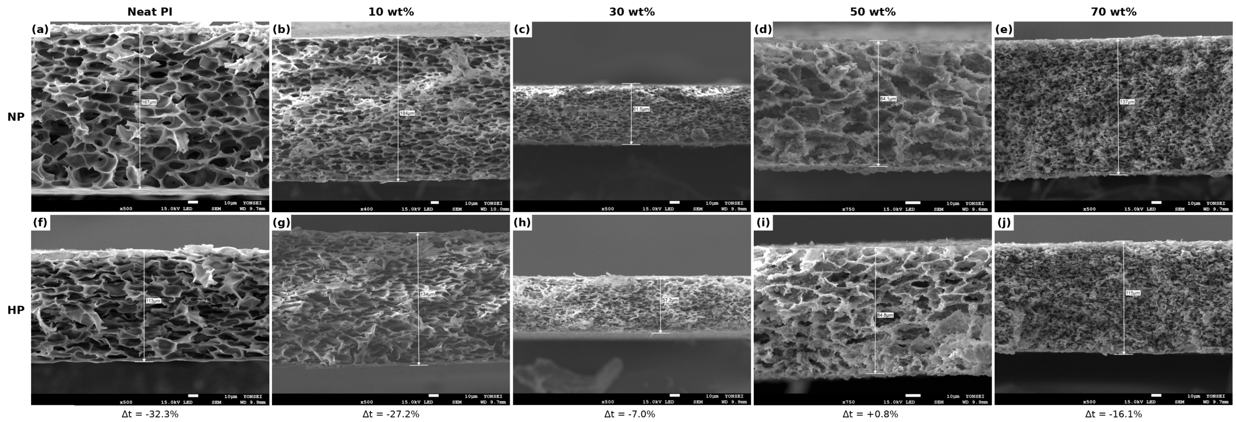

Figure 1: Cross-sectional FE-SEM micrographs of PI/h-BN films at 0–70 wt% h-BN in the non-pressed (a–e, top row) and hot-pressed (f–j, bottom row) states. Δt denotes the film-thickness change upon hot-pressing. Top-view micrographs and EDS elemental analysis are provided in Figure S7 and Table S1.

Hot-pressing collapses part of the open pore network and raises bulk density and tensile strength. At most compositions the HP open porosity is far below the NP value (43.4% vs. 75.4% for neat PI; 29.8% vs. 91.4% at 10 wt%), but at 30 and 50 wt% the HP films show higher intrusion porosities than their NP counterparts (67.2% vs. 48.8%; 64.6% vs. 58.1%). Mercury intrusion measures only the open, connected pore volume. The total porosity from the bulk density, $\phi_{\text{total}} = 1 - \rho/\rho_{\text{solid}}$ (Table 3), exceeds the intrusion value for six of the seven films with measured ρ, by up to 31 percentage points (NP 30 wt%: 79.8% vs. 48.8%), and the difference is a measure of closed porosity. Notably, ϕ_{total} falls from NP to HP at every composition (30 wt%: 79.8 $\rightarrow$ 71.6%; 50 wt%: 70.6 $\rightarrow$ 61.3%; 70 wt%: 72.8 $\rightarrow$ 72.1%). The reversed ordering of the intrusion values at 30 and 50 wt% therefore reflects a change in pore connectivity, in which compaction opens previously closed pores to intrusion, not an increase in pore volume. Both quantities are listed in Table 3; the choice of porosity input to Eq. 3 is discussed in Section 4.2. The SEM cross-sections agree: thickness falls by 32.3%, 27.2%, and 16.1% at 0, 10, and 70 wt%, but changes by only -7.0% and $+0.8\%$ at 30 and 50 wt%, consistent with the resistance of the h-BN network to compaction at these loadings.

EDS (Table S1) shows the boron content rising with nominal loading (8.0 to 29.3 wt% B from 10 to 70 wt% h-BN, NP series), preserved after hot-pressing, and within $\pm$5 wt% (absolute) of the theoretical values at $\geq$10 wt%; the apparent 5.4 wt% B in the neat film is the quantification limit of EDS for light elements and is treated as background. Spot-to-spot variation below 1 wt% B at 70 wt% confirms that the sonication-assisted NIPS protocol gives a homogeneous dispersion.

4.2. Dielectric constant

Figure 2a compares the measured dielectric constants with the three-phase LN predictions for all ten films (parity plot in Figure S2). Every film lies far below dense PI (∼3.5): the NP films span $\varepsilon_r = 1.28$–1.76 at 100 kHz and the HP films 1.62–2.18, so NIPS porosity dominates the dielectric response, and hot-pressing raises ε_r by 0.18–0.42 at every composition through pore collapse. In the NP series ε_r falls monotonically with h-BN content (1.76 → 1.28). Because $\varepsilon_{\mathrm{BN}}$ (∼6) exceeds $\varepsilon_{\mathrm{PI}}$ (∼3.5) and the open porosity does not rise over this range, no mixing rule reproduces this trend (Section S3.5); the model predictions, evaluated at each film's open porosity, follow the porosity input rather than the composition.

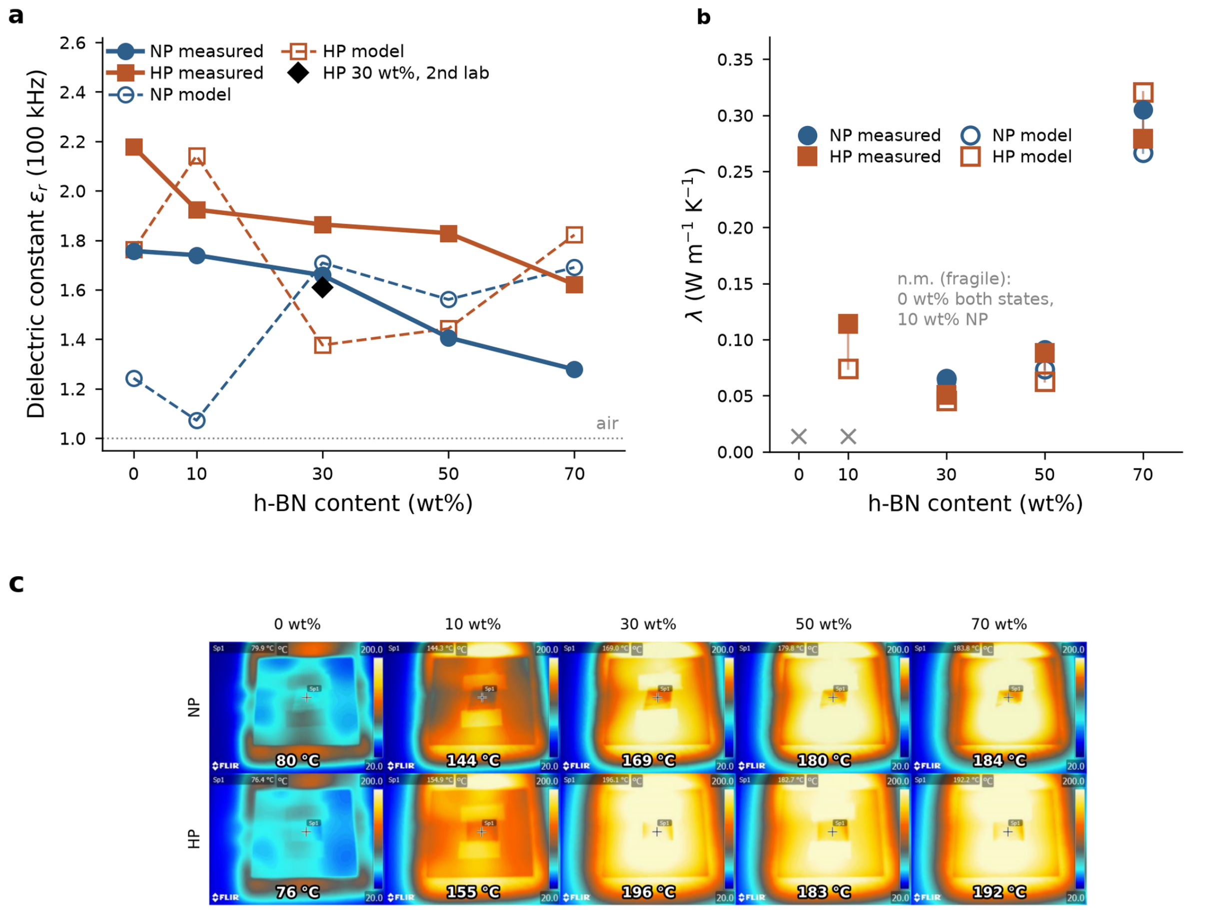

Figure 2: Measured dielectric and thermal properties of PI/h-BN films ($n = 1$ per condition). (a) Dielectric constant at 100 kHz vs. h-BN content for NP and HP films; filled symbols, measured; open symbols joined by dashed lines, three-phase LN predictions at each film's measured open porosity; black diamond, independent measurement of the HP 30 wt% film (KOPTRI, ASTM D150), not used in fitting. (b) Thermal conductivity vs. h-BN content; open symbols, single-parameter LN predictions ($\lambda_{\mathrm{BN,eff}}$ fitted; $A = 5.5$, $\phi_{\max} = 0.60$ fixed), joined to the measured value by a tie-line; crosses mark compositions that could not be measured (film fragility). (c) Infrared thermography of all ten films on a 200 °C stage after [5] min equilibration; annotated values are uncorrected radiometric spot temperatures and are comparative, not absolute. Parity plot and porosity-sensitivity curves are given in Figures S2 and S3.

The model returns RMSE = 0.43 (NP) and 0.36 (HP), larger than the standard deviation of the measured ε_r across compositions (0.19 NP, 0.18 HP), so the coefficient of determination is negative ($R^2 = -3.9$ NP, -3.0 HP; -1.6 for the 30–70 wt% subset). The calibrated model does not predict ε_r for an individual film better than the data mean, and we use it accordingly: as a mechanistic description of how porosity and pore geometry enter ε_r, and as a generator of design maps whose absolute levels carry an uncertainty of the order of the RMSE. Part of the residual lies in the porosity input. The fits use the open porosity, which is available for all ten films; ε_r and λ respond to the total air fraction regardless of connectivity, so ϕ_{total} is the physically appropriate input, but it exists only for the seven films with measured density and differs from the open porosity by up to 31 percentage points (Section 4.1). The largest residuals are at 0, 10, and 70 wt% in the NP series: underprediction by 29–38% at 0–10 wt% and overprediction by 32% at 70 wt% (Table 2).

Two mechanisms account for the underprediction at low loading. First, the NP films at 0 and 10 wt% have porosities of 75.4% and 91.4%. Above $\phi_p \approx 0.75$ the sponge passes from moderate connectivity, where the mean-field $A_{\mathrm{pore}} = 0.08$ is adequate, to a nearly open framework in which the PI skeleton itself becomes discontinuous and long-range connectivity and wall thinning, not local mixing, set the response. Second, at 10 wt% and 91.4% porosity the h-BN volume fraction in the film is only $\phi_{\mathrm{BN,film}} = 0.006$, so the film should behave as a PI foam; yet the opposite limit, isolated PI inclusions in air, gives $\varepsilon_r \approx 1+3\phi_s(\varepsilon_{\mathrm{PI}}-1)/(\varepsilon_{\mathrm{PI}}+2) \approx 1.12$ at $\phi_s \approx 0.087$, also far below the measured 1.74. The real skeleton is evidently more connected than either idealized topology, a partially inverted structure that no mean-field picture captures. These limitations are confined to the high-porosity end and do not affect the conclusions at moderate porosity: for the NP films of 30–70 wt% the RMSE is 0.25.

An independent measurement of the HP 30 wt% film at KOPTRI (ASTM D150) gives $\varepsilon_r = 1.610$, against 1.864 in-house (-13.6%) and a model value of 1.376 at the open porosity of 67.2% (Figure 2a, black diamond; Table 2); the three values disagree pairwise at the 14% level. The inter-laboratory difference estimates the reproducibility of ε_r for a porous film with contact electrodes: an air gap between a rough surface and the electrode acts as a series capacitance that lowers the apparent ε_r by an amount that depends on surface texture, and therefore on composition and processing state. This effect is a plausible experimental contributor to the negative R^2 and to the composition trend of the NP series, and re-measurement with sputtered or painted electrodes would settle it. The model residual at 30 wt% HP (-26%), however, exceeds the inter-laboratory spread and is a model limitation rather than a measurement artefact.

4.3. Thermal conductivity

Figure 2b shows the thermal conductivities, and Table 3 the underlying density, specific heat, and diffusivity together with the open and total porosities. The neat films and the NP 10 wt% film were too fragile to measure (porosities up to 91.4%), leaving seven data points. The single-parameter LN calibration reproduces all of them ($\mathrm{RMSE_{NP}} = 0.027$, $\mathrm{RMSE_{HP}} = 0.030\,\mathrm{W\,m^{-1}\,K^{-1}}$): λ is nearly constant from 30 to 50 wt% and rises sharply at 70 wt%. In the model the rise comes from the packing term ψ in Eq. 1, since

Table 2: Experimental vs. predicted dielectric constants (three-phase Lewis–Nielsen model). KOPTRI: independent inter-laboratory measurement of the HP 30 wt% film (ASTM D150; not used in fitting).

h-BN (wt%)	$\varepsilon_{r,\mathrm{NP}}$ (Exp.)	$\varepsilon_{r,\mathrm{NP}}$ (LN)	Δ (%)	$\varepsilon_{r,\mathrm{HP}}$ (Exp.)	$\varepsilon_{r,\mathrm{HP}}$ (LN)	Δ (%)	KOPTRI (HP)
0	1.757	1.242	−29.3	2.178	1.760	−19.2	—
10	1.740	1.073	−38.4	1.924	2.139	+11.2	—
30	1.659	1.707	+2.9	1.864	1.376	−26.2	1.610*
50	1.407	1.560	+10.8	1.829	1.442	−21.2	—
70	1.278	1.689	+32.1	1.622	1.819	+12.2	—

$\Delta = (\varepsilon_{r,\mathrm{LN}} - \varepsilon_{r,\mathrm{Exp}})/\varepsilon_{r,\mathrm{Exp}} \times 100\%$. LN predictions are evaluated at each film's open porosity. *Inter-laboratory deviation from the in-house HP value (1.864) is −13.6%; the model prediction at the open porosity of 67.2% is 1.376 (Section 4.2).

at $\phi_{\mathrm{BN}} = 0.591 \approx 0.99\,\phi_{\max}$ the composite is at the platelet packing limit, which is the LN representation of imminent particle–particle contact.

Table 3: Porosity and thermophysical data underlying the thermal-conductivity determination ($\lambda = \alpha\rho C_p$; laser flash at 25 °C). ϕ_{open}: mercury intrusion; $\phi_{\mathrm{total}} = 1 - \rho/\rho_{\mathrm{solid}}$ with $\rho_{\mathrm{PI}} = 1.42$ and $\rho_{\mathrm{BN}} = 2.29\,\mathrm{g\,cm^{-3}}$.

h-BN (wt%)	State	ϕ_{open} (%)	ϕ_{total} (%)	ρ ($\mathrm{g\,cm^{-3}}$)	C_p ($\mathrm{J\,g^{-1}\,K^{-1}}$)	α ($\mathrm{mm^2\,s^{-1}}$)	λ ($\mathrm{W\,m^{-1}\,K^{-1}}$)
0	NP	75.4	—	—	—	—	—
0	HP	43.4	—	—	—	—	—
10	NP	91.4	—	—	—	—	—
10	HP	29.8	40.4	0.879	0.903	0.143	0.114
30	NP	48.8	79.8	0.323	0.853	0.230	0.063
30	HP	67.2	71.6	0.455	0.863	0.131	0.051
50	NP	58.1	70.6	0.515	0.875	0.202	0.091
50	HP	64.6	61.3	0.678	0.885	0.147	0.088
70	NP	54.2	72.8	0.526	0.838	0.693	0.305
70	HP	48.8	72.1	0.539	0.846	0.613	0.279

Neat PI (both states) and NP 10 wt% could not be measured by laser flash (film fragility), and ρ was not determined for these films. The LN fits use ϕ_{open} as the porosity input (Section 4.2).

The fitted $\lambda_{\mathrm{BN,eff}} = 6.81\,\mathrm{W\,m^{-1}\,K^{-1}}$ lies well below the through-plane conductivity of bulk h-BN (∼[30] $\mathrm{W\,m^{-1}\,K^{-1}}$ [14]). We attribute the reduction mainly to Kapitza resistance at the h-BN–PI interface: at 70 nm the surface-to-volume ratio is about $30\,\mathrm{m^2\,g^{-1}}$, so interfacial phonon scattering limits heat transfer through each particle. Molecular-dynamics estimates of the interfacial conductance of h-BN–polymer interfaces ($G \sim$ 10–100 $\mathrm{MW\,m^{-2}\,K^{-1}}$) translate at 70 nm into effective particle conductivities of 1–7 $\mathrm{W\,m^{-1}\,K^{-1}}$ [30], consistent in magnitude with the fitted value. Two residuals carry physical information, with the caveat that both are comparable to the ∼10% propagated uncertainty in λ (Section 2.5). First, the NP 70 wt% film exceeds the prediction (0.305 vs. 0.263 $\mathrm{W\,m^{-1}\,K^{-1}}$, +16%), consistent with incipient h-BN–h-BN contact beyond the mean-field description. Second, hot-pressing at 70 wt% lowers λ from 0.305 to 0.279 $\mathrm{W\,m^{-1}\,K^{-1}}$ although the open porosity falls (54.2 → 48.8%) and the total porosity is nearly unchanged (72.8 → 72.1%); the model predicts a rise to 0.315. Compaction evidently disrupts part of the h-BN contact network formed during NIPS, sacrificing percolative pathways even as the solid fraction increases. At fixed composition the two properties remain coupled through porosity (Figure S3): at 50 wt%, over the accessible 30–65% porosity range, ε_r spans 1.43–2.31 and λ 0.06–0.16 $\mathrm{W\,m^{-1}\,K^{-1}}$.

4.4. Tensile strength and coefficient of thermal expansion

Hot-pressing raises the tensile strength by 19–95% at every composition (Figure 3a; Table 4), as expected for pore collapse under compressive load. The NP strength is non-monotonic, with a maximum at 30 wt% (25.4 MPa) and minima at 10 wt% (14.9 MPa) and 50 wt% (12.5 MPa), and follows the porosities at these compositions (91.4% and 58.1%) inversely. The LN + Gibson–Ashby model reproduces this trend qualitatively.

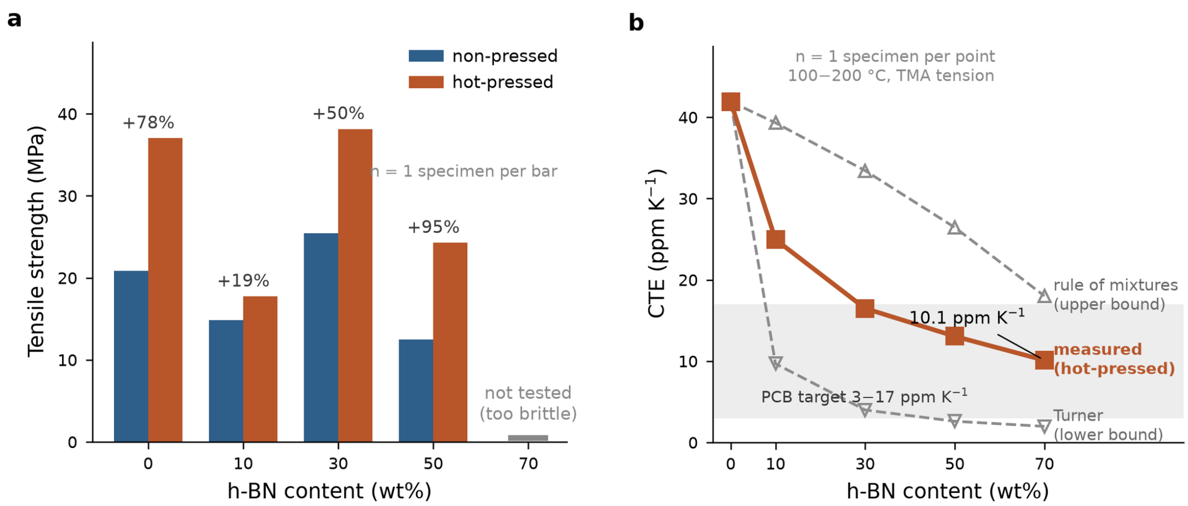

Figure 3: Mechanical and thermal-expansion behavior of PI/h-BN films. (a) Tensile strength vs. h-BN content for NP and HP films ($n = 1$ specimen per bar). Percentage values indicate the NP→HP change at each composition; the 70 wt% slot carries a baseline stub because those films were too brittle to test, not because the strength is zero. (b) CTE vs. h-BN content for HP films ($n = 1$ per point; TMA, tension mode, 100–200 °C): measured values against the Turner lower bound and the rule-of-mixtures (ROM) upper bound, both computed with $\alpha_{\mathrm{PI}} = 41.92\,\mathrm{ppm\,K^{-1}}$ (measured). The PCB substrate target zone (3–17 $\mathrm{ppm\,K^{-1}}$) is shaded. The measured CTE lies between the two bounds because the discontinuous load-path geometry of the porous sponge gives only partial elastic constraint (Section 3.3).

The measured CTE falls from 24.96 to 10.11 $\mathrm{ppm\,K^{-1}}$ between 10 and 70 wt% (Figure 3b). The Turner model, with $\alpha_{\mathrm{PI}} = 41.92\,\mathrm{ppm\,K^{-1}}$ from the neat PI HP film, reproduces the neat value but underpredicts by 61–80% at every loading, and the ROM overpredicts because it ignores elastic constraint. The data lie between the two bounds, as expected for partial constraint in a sponge whose platelets sit in discontinuous PI ligaments rather than in a continuous dense matrix. Dense-composite CTE models therefore reach their limit for sponge materials, and an open-cell foam mechanics treatment is a natural next step. At 70 wt% (HP) the CTE of 10.1 $\mathrm{ppm\,K^{-1}}$ lies within the PCB substrate window of 3–17 $\mathrm{ppm\,K^{-1}}$. **[CTE was measured on the HP films only; the NP films at ≥50 wt% were too fragile for TMA clamping.]**

4.5. Design maps for simultaneous property optimization

Figure 4a,b maps the model ε_r and λ over the full design space (0–70 wt% h-BN, 0–95% porosity), with the measured NP and HP states overlaid and arrows marking the NP→HP trajectory at each composition.

Three design rules follow. First, $\varepsilon_r < 1.5$ requires a porosity above ∼57% at low loading and ∼63% at 70 wt%, a range that NIPS reaches in both the NP and moderately pressed states. Second, λ is gated by

Table 4: Tensile strength and CTE data for PI/h-BN composite films.

h-BN (wt%)	NP strength (MPa)	HP strength (MPa)	NP→HP (%)	HP CTE ($\mathrm{ppm\,K^{-1}}$)	Turner ($\mathrm{ppm\,K^{-1}}$)	ROM ($\mathrm{ppm\,K^{-1}}$)
0	20.85	37.02	+77.6	41.92	41.92	41.92
10	14.86	17.74	+19.4	24.96	9.69	39.31
30	25.43	38.15	+50.0	16.49	4.00	33.43
50	12.47	24.28	+94.7	13.06	2.61	26.45
70	—*	—*	—	10.11	1.98	18.02

*Not tested (film brittleness at 70 wt% h-BN). Turner and ROM use $\alpha_{\mathrm{PI}} = 41.92\,\mathrm{ppm\,K^{-1}}$ (measured neat porous PI, HP), $E_{\mathrm{PI}} = 3.5\,\mathrm{GPa}$, $E_{\mathrm{BN}} = 200\,\mathrm{GPa}$, $\alpha_{\mathrm{BN}} = 1.5\,\mathrm{ppm\,K^{-1}}$, solid-phase volume fractions.

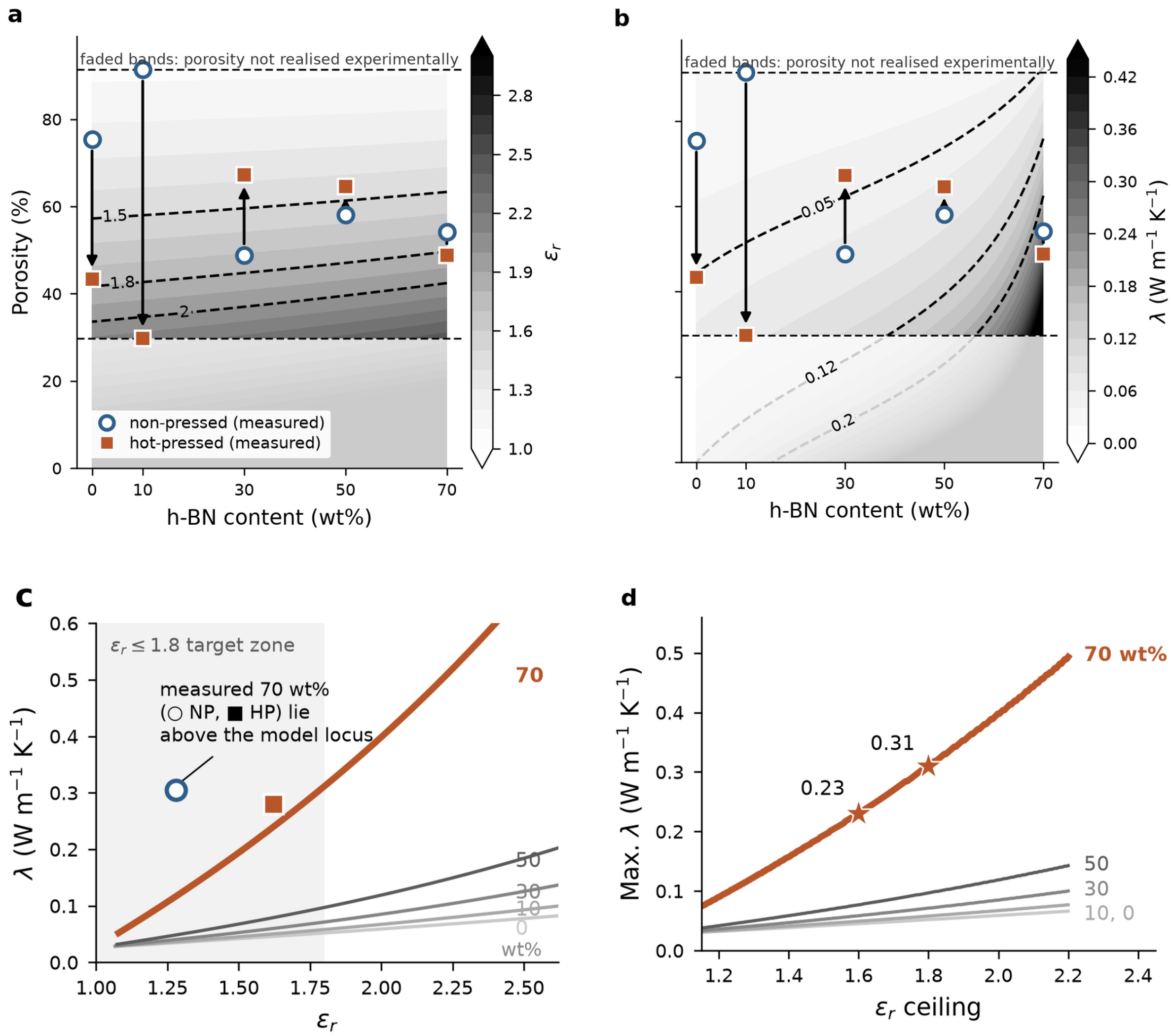

Figure 4: Model-guided design (three-phase LN model, calibration of Table 1). (a, b) Design maps of ε_r and λ vs. h-BN content and porosity; measured NP (open circles) and HP (filled squares) states overlaid, arrows indicating the NP→HP trajectory; faded bands above 91% and below 30% porosity lie outside the range realized by any film. (c) ε_r–λ loci for each loading with porosity swept along each curve; the measured 70 wt% films lie above their own locus. (d) Maximum λ attainable under a given ε_r ceiling with porosity optimized at each point; stars mark the $\varepsilon_r \leq 1.6$ and $\varepsilon_r \leq 1.8$ operating points at 70 wt%. The absolute levels of panels (a)–(d) inherit the dielectric uncertainty quantified in Section 4.2. Feasible-window and porosity-optimum panels are given in Figures S4 and S5.

loading: $\lambda > 0.20\,\mathrm{W\,m^{-1}\,K^{-1}}$ is out of reach below ∼55 wt% at any practical porosity, needs a porosity below ∼35% at 60 wt%, but is available up to ∼62% porosity at 70 wt%, because the packing-limit amplification at high loading relaxes the porosity constraint. Third, hot-pressing moves a film almost vertically on the maps, so composition must be chosen first and porosity tuned second; at 70 wt% both the NP and HP films already lie in the region favorable for both properties.

4.6. Shape-factor sensitivity

Figure S1 examines the sensitivity of the predictions to the h-BN shape factor over $A = 0.3$–6.0, from needle-like rods to aligned platelets. For the dielectric model (Figure S1a) the curves are nearly indistinguishable: ε_r is set by the pore term (step 2, Eq. 3), not by A_{BN}, so in porous NIPS films the particle geometry has little influence and only the porosity and pore architecture (A_{pore}) matter. This justifies treating the fitted dielectric A_{BN} as weakly determined (Table 1) and confirms that the residuals in Table 2 reflect mean-field limitations rather than parameter uncertainty.

The thermal model (Figure S1b) is more sensitive to A above 30 wt%, which is why we fixed A at its literature platelet value rather than fitting it (Section 3.4). At the NP mean porosity of 65.6%, every curve from $A = 0.3$ to 6.0 stays below the measured NP 70 wt% value (0.16–0.18 vs. $0.305\,\mathrm{W\,m^{-1}\,K^{-1}}$), so the 70 wt% enhancement comes from the proximity to maximum packing and incipient particle contact, not from any single-particle shape factor. Over 30–50 wt% the curves converge within $\pm 0.02\,\mathrm{W\,m^{-1}\,K^{-1}}$, and the mean-field model is adequate.

4.7. Property trade-offs and design guidance

For each loading the calibrated model traces a locus in ε_r–λ space as porosity is swept (Figure 4c); the 70 wt% HP film ($\varepsilon_r = 1.62$, $\lambda = 0.28\,\mathrm{W\,m^{-1}\,K^{-1}}$) already lies in the target zone. Three points follow. Along any locus, lowering porosity raises ε_r and λ together; the two are coupled through the same structural variable, and every porosity is a position on a trade-off curve. Raising the loading shifts the whole locus to higher λ at a given ε_r, so composition is the lever for thermal performance at a fixed dielectric specification. The 70 wt% locus runs through the most favorable region: the NP film at 54.2% open porosity ($\varepsilon_r = 1.28$, $\lambda = 0.31\,\mathrm{W\,m^{-1}\,K^{-1}}$) meets all three target levels defined below, and the HP film at 48.8% ($\varepsilon_r = 1.62$, $\lambda = 0.28\,\mathrm{W\,m^{-1}\,K^{-1}}$) meets the Basic and Advanced levels.

4.8. Model-guided design: performance limits and processing prescriptions

The calibrated model can be interrogated over the full (loading, porosity) space for performance limits and for the processing conditions that reach a given specification (Figure 4c,d); the absolute levels of these surfaces inherit the dielectric uncertainty of Section 4.2.

We define three specifications of increasing difficulty: Basic ($\varepsilon_r < 2.0$, $\lambda > 0.05\,\mathrm{W\,m^{-1}\,K^{-1}}$), Advanced ($\varepsilon_r < 1.8$, $\lambda > 0.08\,\mathrm{W\,m^{-1}\,K^{-1}}$), and Future ($\varepsilon_r < 1.6$, $\lambda > 0.12\,\mathrm{W\,m^{-1}\,K^{-1}}$). The porosity window that satisfies each is shown in Figure S4. The Basic window is open at every composition (neat porous PI qualifies at $\sim$34–44% porosity, since $\lambda_{\mathrm{PI}} = 0.12\,\mathrm{W\,m^{-1}\,K^{-1}}$ clears the threshold at moderate porosity); the Advanced window opens at $\geq$40 wt%; and the Future window opens only at $\geq$64 wt%, spanning 58–76% porosity at 70 wt%. Experimentally, the 70 wt% NP film ($\varepsilon_r = 1.28$, $\lambda = 0.31\,\mathrm{W\,m^{-1}\,K^{-1}}$) meets the Future specification and the HP film meets the Advanced one. The model is conservative in ε_r at this composition (1.69 predicted vs. 1.28 measured), so the films outperform the model envelope and the windows should be read as sufficient, not necessary, conditions.

Figure 4c overlays the loci of all loadings with the Pareto frontier computed over the full grid; the frontier coincides with the 70 wt% locus across the accessible range. Because the dielectric residuals exceed the spread of the data and the measured 70 wt% films depart from their locus ($\varepsilon_r = 1.28$ vs. 1.69 predicted, NP), we read this as model-level consistency with the experimental finding that 70 wt% is the most favorable composition tested, not as a validated frontier of the material system beyond the measured range. Figure 4d answers the practical question of the maximum λ at a given ε_r ceiling: at 70 wt% the envelope reaches $0.23\,\mathrm{W\,m^{-1}\,K^{-1}}$ at $\varepsilon_r \leq 1.6$ (58% porosity), $0.31\,\mathrm{W\,m^{-1}\,K^{-1}}$ at $\varepsilon_r \leq 1.8$ (50%), and $0.39\,\mathrm{W\,m^{-1}\,K^{-1}}$ at $\varepsilon_r \leq 2.0$ (42%). The measured 70 wt% films (0.305 NP, $0.279\,\mathrm{W\,m^{-1}\,K^{-1}}$ HP) lie on or slightly above the $\varepsilon_r \leq 1.8$ envelope, consistent with the surface in the region probed.

The surface reduces to simple prescriptions. The porosity that maximizes λ under a given ε_r ceiling is nearly independent of composition (42–50% for $\varepsilon_r \leq 1.8$, 52–58% for $\varepsilon_r \leq 1.6$; Figure S5), so the dielectric ceiling sets the porosity target and the loading sets the λ obtainable there. In particular, $\varepsilon_r < 1.7$ with $\lambda > 100\,\mathrm{mW\,m^{-1}\,K^{-1}}$ needs $\geq$60 wt% at 53–54% porosity; $\varepsilon_r < 1.6$ with $\lambda > 120\,\mathrm{mW\,m^{-1}\,K^{-1}}$ needs 70 wt% at 58%; and $\varepsilon_r < 1.5$ with $\lambda > 150\,\mathrm{mW\,m^{-1}\,K^{-1}}$ is feasible only at 70 wt% and 63%. Since the as-fabricated 70 wt% NP film sits at 54.2%, these targets are reached by mild or partial hot-pressing. Full densification is neither required nor desirable: it would push ε_r past the ceilings and, as Section 4.3 shows, risks breaking the h-BN contact network.

5. Conclusions

We fabricated porous PI/h-BN films by NIPS over 0–70 wt% h-BN, characterized them in the non-pressed and hot-pressed states, and described the full dataset with a calibrated three-phase Lewis–Nielsen framework. NIPS porosity is the dominant determinant of the dielectric constant: all films lie at $\varepsilon_r = 1.28$–2.18 (100 kHz), far below dense PI ($\sim$3.5), and hot-pressing raises ε_r by 0.18–0.42 through pore collapse. The fitted pore shape factor $A_{\mathrm{pore}} = 0.08$ quantifies the low field-shielding efficiency of the bicontinuous NIPS sponge relative to isolated spherical pores ($A = 1.5$) and provides a model-based fingerprint that distinguishes NIPS from Pickering-emulsion-templated porous PI/h-BN films. Because the dielectric residuals exceed the spread of the data, the model serves as a mechanistic and design tool rather than as a quantitative predictor of ε_r for individual films.

A single-parameter thermal calibration ($\lambda_{\mathrm{BN,eff}} = 6.81\,\mathrm{W\,m^{-1}\,K^{-1}}$, with platelet shape factor and packing fraction fixed at literature values) reproduces the conductivity data with RMSE $\leq 0.03\,\mathrm{W\,m^{-1}\,K^{-1}}$.

The reduction of $\lambda_{\mathrm{BN,eff}}$ well below bulk h-BN quantifies the interfacial (Kapitza) resistance at 70 nm particle size, and the sharp rise in λ at 70 wt% emerges from the approach to maximum platelet packing ($\phi_{\mathrm{BN}} = 0.59 \approx 0.99\,\phi_{\mathrm{max}}$) without an additional percolation parameter. Hot-pressing at 70 wt% lowers λ from 0.305 to 0.279 $\mathrm{W\,m^{-1}\,K^{-1}}$, opposite to the model trend, indicating that compaction partially disrupts the h-BN contact network formed during NIPS. The measured CTE lies between the Turner and rule-of-mixtures bounds because the sponge breaks the continuous load-path assumption of dense-composite models, and reaches 10.1 $\mathrm{ppm\,K^{-1}}$ at 70 wt% (HP), within the PCB substrate window.

Within the compositions tested, 70 wt% h-BN gives the most favorable ε_r–λ combination, and the calibrated model is consistent with this result: the 70 wt% locus lies above every other composition across the accessible porosity range, and the model prescribes a porosity window of 50–63% at 70 wt%, reachable from the as-fabricated 54% by mild hot-pressing. These results establish the pore shape factor as a microstructural design parameter for porous ceramic-filled polymer films and provide a calibrated basis for selecting composition and porosity in NIPS-derived PI/h-BN packaging films.

CRediT authorship contribution statement

Jinyoung Kim: Conceptualization, Methodology, Investigation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing. **[Add co-authors and their roles if applicable.]**

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All experimental data underlying the figures are provided in the Supplementary Material and in the accompanying raw-data workbook. The Python implementation of the three-phase Lewis–Nielsen model, including the parameter-estimation protocol, is listed in Section S5 of the Supplementary Material and archived in a public repository (**[https://doi.org/10.5281/zenodo.XXXXXXX]**).

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